Report Profile likelihood based identifiability analysis

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October 2026

1 Parameter Estimation

We consider biological models, described by systems of ODEs:

$$\dot{\mathbf{x}}(t, \boldsymbol{\theta}_{\text{dyn}}) = f(\mathbf{x}(t), \boldsymbol{\theta}_{\text{dyn}}, \mathbf{u}(t)), \tag{1}$$

where $\mathbf{x} : \mathbb{R} \to \mathbb{R}^n : t \mapsto \mathbf{x}(t)$ are the internal states or dynamical variables of the system, corresponding to for example molecular species such as hormones, proteins or mRNA. $\boldsymbol{\theta}_{\text{dyn}} \in \mathbb{R}^{p_1}$ denotes the set of dynamical parameters such as rate constants and initial concentrations of the dynamical variables, $\mathbf{x}(0, \boldsymbol{\theta}_{\text{dyn}}) = \mathbf{x}_0(\boldsymbol{\theta}_{\text{dyn}})$.

The equations are then solved with numerical integration. Given time-series data at timepoints $t_i \in \{t_1, \ldots, t_d\}$, the goal is then to compare the system dynamics to this data. For a given parameter $\boldsymbol{\theta}_{\text{dyn}}$, the dynamic variables are mapped to observables

$$\mathbf{y}(t_i, \boldsymbol{\theta}) = \mathbf{g}(\mathbf{x}(t_i, \boldsymbol{\theta}_{\text{dyn}}), \boldsymbol{\theta}_{\text{obs}}) \tag{2}$$

The observables $\mathbf{y} : \mathbb{R} \to \mathbb{R}^m : t \mapsto \mathbf{y}(t)$ are the quantities that can be measured in experiments and can be directly compared to the experimental data.

The likelihood $L(\mathbf{Y}|\boldsymbol{\theta})$ is the joint probability distribution of observing the data $\mathbf{Y}$, given a set of parameters $\boldsymbol{\theta}$. Assuming that the measurement noise is statistically independent and additive, the expression is factorized as follows:

$$\begin{aligned} L(\mathbf{Y}|\boldsymbol{\theta}) &= \prod_{k,i} p(Y_k(t_i)|\boldsymbol{\theta}) \\ &= \prod_{k=1}^{m} \prod_{i=1}^{d} \frac{1}{\sqrt{2\pi\sigma_{ki}^2}} \exp\left(-\frac{1}{2\sigma_{ki}^2}\left(Y_{ki} - y_k(t_i, \boldsymbol{\theta})\right)^2\right). \end{aligned}$$

The maximum likelihood estimate (MLE) of the parameters $\boldsymbol{\theta}$ is obtained by maximizing the likelihood. However, it is numerically more efficient to minimize $-2 \cdot \log L(\boldsymbol{Y}|\boldsymbol{\theta})$. Its expression is

$$-2 \cdot \log L(\mathbf{Y} \mid \boldsymbol{\theta}) = \sum_{k=1}^{m} \sum_{i=1}^{d} \left(\log\left(2\pi\sigma_{ki}^2\right) + \frac{\left(Y_{ki} - y_k(t_i, \boldsymbol{\theta})\right)^2}{\sigma_{ki}^2} \right) \tag{3}$$

which is, except for a constant offset, equivalent to the weighted residual sum of squares:

$$\sum_{k,i} \frac{(Y_{ki} - y_k(t_i, \boldsymbol{\theta})^2}{\sigma_{ki}^2}.$$

In this work the expression $-2 \cdot \log L(\boldsymbol{\theta})$ will also be referred to as the objective function.

In conclusion, the *maximum likelihood estimate* $\boldsymbol{\theta}_{MLE}$ corresponds to minimizing the negative log likelihood

$$\boldsymbol{\theta}_{MLE} = \arg\min_{\boldsymbol{\theta}} \left(-2 \cdot L(\mathbf{Y} \mid \boldsymbol{\theta})\right).$$

The log likelihood is always defined over $\log 10$ scale of the parameter search space.

2 The Profile Likelihood and Practical identifiability

The profile likelihood [2] for an individual parameter $\theta_i \in \boldsymbol{\theta}$ is defined as

$$PL_i(p) = \max_{\theta \in \{\theta | \theta_i = p\}} \log L(\boldsymbol{\theta}), \tag{4}$$

which can be rewritten (using the notation in equation (**??**)) as

$$\chi^2_{PL_i}(p) \coloneqq -2 \cdot PL_i(p) = \min_{\theta \in \{\theta | \theta_i = p\}} \chi^2(\boldsymbol{\theta}), \tag{5}$$

This function allows for exploring the parameter space for each parameter individually. As mentioned before, logarithmic parameter space is considered here. The remaining parameters $\theta_{j \neq i}$ are treated as nuisance parameters that are allowed to vary to re-optimize the objective function while θ_i is kept fixed. This implies that the parameter space is explored for θ_i in the direction of the least increase of the objective function.

With this approach, each parameter in $\boldsymbol{\theta}$ can be classified as **identifiable, practically non-identifiable**, or **structurally nonidentifiable**. Structural nonidentifiability arises when a subset of parameters $\boldsymbol{\theta}_{\text{sub}} \subset \boldsymbol{\theta}$ can be varied without altering the model output $\boldsymbol{y}(t)$. Consequently, the objective function remains constant at its minimum value along this parameter trajectory.

The profile likelihood ratio for a parameter subset of interest $\boldsymbol{\theta}_1 \subset \boldsymbol{\theta}$ and nuisance parameters $\boldsymbol{\theta}_2 \subset \boldsymbol{\theta}$ is defined as [1]

$$R(\boldsymbol{\theta}_1) = \max_{\boldsymbol{\theta}_2} \left[\frac{L(\boldsymbol{\theta}_1, \boldsymbol{\theta}_2)}{L(\boldsymbol{\theta}_{MLE})} \right]. \tag{6}$$

Wilks' theorem [3] implies that under certain regularity conditions, the profile likelihood ratio statistic follows a chi-square distribution with $k_1 = \text{length}(\boldsymbol{\theta}_1)$ degrees of freedom, as the number of data samples n converges. This can be referred to as the *asymptotic* result.

A likelihood-based confidence interval $\text{CI}_{i,\alpha}$ (at significance level α) for an individual parameter θ_i is defined as

$$\text{CI}_{i,\alpha}(\mathbf{Y}) = \left\{ p \mid \chi^2_{PL_i}(p) - \chi^2(\boldsymbol{\theta}_{MLE}) < \Delta_{(1-\alpha)} \right\}, \tag{7}$$

where the expression $\chi^2_{PL_i}(p) - \chi^2(\boldsymbol{\theta}_{MLE})$ directly corresponds to the profile likelihood ratio $-2 \log R(\theta_i)$ with one parameter of interest. Then, in the asymptotic case, $\Delta_{(1-\alpha)}$ is the $(1-\alpha)$ percentile of the chi-square distribution with one degree of freedom ($df = 1$). However, in biological modeling, experimental data is often scarce and the necessary conditions may not hold such that the asymptotic value for $\Delta_{(1-\alpha)}$ is a good approximation. The *bootstrap* method can be applied to the profile likelihood ratio statistic to estimate the threshold value.

Equation (7) can be rewritten as

$$\text{CI}_{i,\alpha}(\mathbf{Y}) = \left\{ p \mid \chi^2_{PL_i}(p) < \chi^2(\boldsymbol{\theta}_{MLE}) + \triangle_{(1-\alpha)} \right\}. \tag{8}$$

The profile likelihood for a structurally unidentifiable parameter θ_i is flat and constant at the minimum objective value. it never crosses the threshold value $\chi^2(\boldsymbol{\theta}_{MLE}) + \triangle_{(1-\alpha)}$ in equation (8). Hence, the likelihood-based confidence interval is *infinite* in both the increasing and decreasing directions of the parameter of interest. On the contrary, the profile likelihood for an identifiable parameter manifests in a *finite* confidence interval.

A parameter that is structurally identifiable can still be practically non-identifiable. **Practical non-identifiability** occurs when the quality and amount of experimental data is *insufficient*. A parameter is defined to be practically unidentifiable if its likelihood-based confidence region (equation (8)) is infinitely extended in increasing or decreasing direction, although the likelihood has a unique minimum for this parameter [2].

3 Results

We applied the workflow to synthetic data generated using the Lotka–Volterra model:

$$\frac{dx}{dt} = \alpha x - \beta y,$$
$$\frac{dy}{dt} = -\gamma y + \delta xy,$$

with the true parameter values $\alpha = 0.52$, $\beta = 0.026$, $\gamma = 0.84$, and $\delta = 0.026$.

The code allows the number of synthetic data points and the measurement uncertainty σ to be specified. It also allows the initial concentrations to be treated as two additional unknown parameters and the observations to include either x, y, or both. In the following, we consider three cases, each using 14 synthetic data points:

1. **Case 1:** Both predator and prey populations are observed.
2. **Case 2:** Only the prey population is observed.
3. **Case 3:** Only the prey population is observed, and the initial population values are unknown.

3.1 Case 1: Two Observables

Table 1 presents the parameter estimates and directional confidence intervals obtained when both populations are observed. The corresponding fitted trajectories are shown in Figure 1.

Table 1: Case 1: Parameter Estimates and Confidence Intervals in Orders of Magnitude. 2 Observables, $\sigma = 0.1$, 14 data points.

Parameter	**Bounds** ($\log_{10}$)	**MLE** ($\log_{10}$)	**Directional Error** ($\log_{10}$)
α	$[-0.30, -0.28]$	-0.2922	$-0.2922^{+0.0122}_{-0.0078}$
β	$[-1.60, -1.58]$	-1.5929	$-1.5929^{+0.0129}_{-0.0071}$
δ	$[-1.59, -1.57]$	-1.5772	$-1.5772^{+0.0072}_{-0.0128}$
γ	$[-0.07, -0.06]$	-0.0665	$-0.0665^{+0.0065}_{-0.0035}$

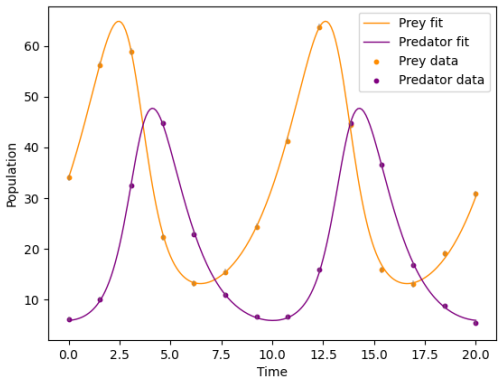


Figure 1: Fitted Lotka–Volterra trajectories for Case 1, with both predator and prey populations observed.

3.2 Case 2: One Observable

In Case 2, only the prey population is observed. The parameter estimates and directional confidence intervals are reported in Table 2, and the corresponding fitted trajectories are shown in Figure 2.

Table 2: Parameter Estimates and Confidence Intervals in Orders of Magnitude. 1 Observable (0,), $\sigma = 0.1$.

Parameter	**Bounds** ($\log_{10}$)	**MLE** ($\log_{10}$)	**Directional Error** ($\log_{10}$)
α	$[-0.29, -0.20]$	-0.2430	$-0.2430^{+0.0430}_{-0.0470}$
β	$[-1.58, -1.38]$	-1.4824	$-1.4824^{+0.1024}_{-0.0976}$
δ	$[-1.68, -1.58]$	-1.6323	$-1.6323^{+0.0523}_{-0.0477}$
γ	$[-0.17, -0.07]$	-0.1229	$-0.1229^{+0.0529}_{-0.0471}$

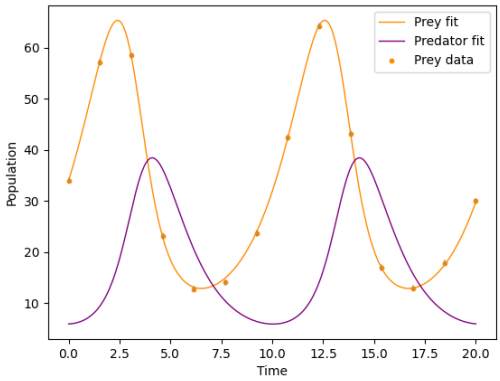


Figure 2: Fitted Lotka–Volterra trajectories for Case 2, with only the prey population observed.

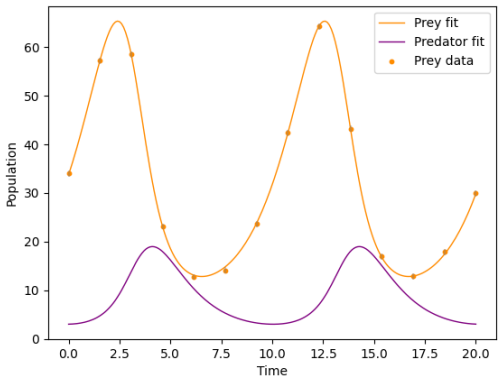


Figure 3

3.3 Case 3: One Observable with Unknown Initial Conditions

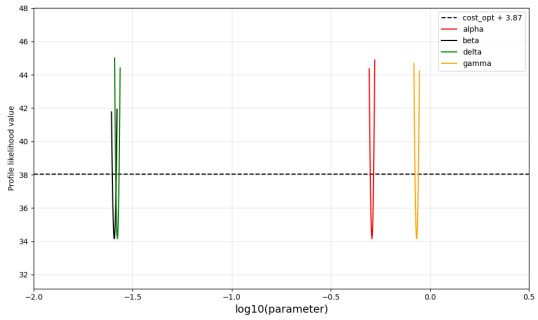
In Case 3, only the prey population is observed, and the initial population values are treated as unknown parameters. Table 3 summarises the resulting parameter estimates and directional confidence intervals. The corresponding fitted trajectories are shown in Figure **??**.

Table 3

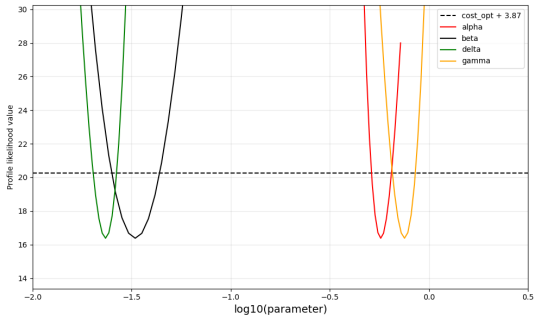
Parameter	**Bounds** ($\log_{10}$)	**MLE** ($\log_{10}$)	**Directional Error** ($\log_{10}$)
α	$[-0.29, -0.17]$	-0.2345	$-0.2345^{+0.0645}_{-0.0555}$
β	$[-2.00, -0.56]$	-1.1726	$-1.1726^{+0.6126}_{-0.8274}$
δ	$[-1.71, -1.58]$	-1.6402	$-1.6402^{+0.0602}_{-0.0698}$
γ	$[-0.20, -0.07]$	-0.1314	$-0.1314^{+0.0614}_{-0.0686}$
prey_0	$[+1.52, +1.54]$	1.5284	$1.5284^{+0.0116}_{-0.0084}$

3.4 Comparison of Profile Likelihoods

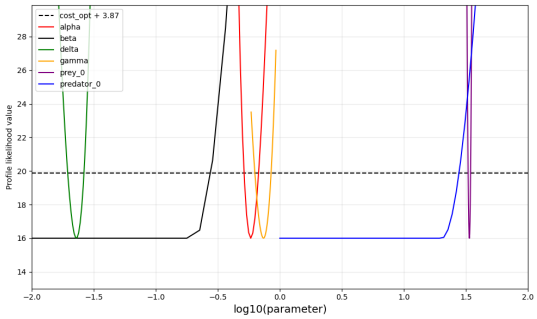
Finally, we compare the profile-likelihood results across the three cases. Figure 4 shows the profiles for Case 1, Case 2, and Case 3, respectively, allowing the effect of observing fewer populations and introducing unknown initial conditions to be assessed.



(a) Case 1: Both populations observed.



(b) Case 2: Prey population observed.



(c) Case 3: Prey population observed, with unknown initial conditions.

Figure 4: Comparison of the profile-likelihood results for the three observation scenarios.

References

[1] William Q Meeker and Luis A Escobar. Teaching about Approximate Confidence Regions Based on Maximum Likelihood Estimation.

[2] A. Raue, C. Kreutz, T. Maiwald, J. Bachmann, M. Schilling, U. Klingmüller, and J. Timmer. Structural and practical identifiability analysis of partially observed dynamical models by exploiting the profile likelihood. *Bioinformatics*, 25(15):1923–1929, August 2009.

[3] S. S. Wilks. The Large-Sample Distribution of the Likelihood Ratio for Testing Composite Hypotheses. *The Annals of Mathematical Statistics*, 9(1):60–62, March 1938.